

End-to-End Endpoint Synthesis on the Literal Fixed-Shift Carrier: Acyclic Proof Certificates, Typed Exponent Transport, and Strict Physical Slack

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

The fixed-shift TPC program now has exact local certificates for physical frames, native grouping, localization, cover, reconnection, determinant energy, and the distinguished zero mode. A remaining risk is not a missing local identity but an invalid global composition: circular hypotheses, repeated physical losses, energy–amplitude confusion, or reuse of one saving in two ledgers. We give an end-to-end typed synthesis theorem that excludes those errors. A proof-carrying route is represented by an acyclic dependency graph with a semantic target firewall and by an occurrence-level physical-event registry. Squarefree truncation, zero-mode content, and physical high-frequency tails receive different namespaces. If every nonsoft literal packet has uniform raw saving $X^{-\sigma_{\text{raw}}}$, the complete soft remainder is $o(X)$, and the physical synthesis cost is $X^{\Lambda_{\text{phys}}+o(1)}$, then the original hard packet is

$$O\left(X^{1-\sigma_{\text{raw}}+\Lambda_{\text{phys}}+o(1)}\right) + o(X).$$

Thus only the strict inequality $\sigma_{\text{raw}} > \Lambda_{\text{phys}}$ closes the route. The MVP1 threshold 1/400 is a specialization. The determinant–zero reserve is proved to be a separate admissibility quantity and cannot be spent at the physical endpoint. The paper is an L0/L1 composition theorem and audit format. It proves no growing coefficient-specific arithmetic estimate.

Keywords: fixed shift; typed exponent ledger; proof DAG; physical synthesis; determinant zero mode; endpoint certificate.

Claim boundary. No input below supplies a Möbius, Liouville, or prime-pair cancellation theorem. A finite exact regression checks the certificate language only. Every growing arithmetic exponent used by an eventual application must be supplied by a separate theorem on the actual fixed- h_0 carrier.

1 Why a final type check is necessary

TPC-113–119 isolate the physical frame, grouping, localization, complete tail, cover, endpoint, and reconnection interfaces [2–8]. TPC-121 and TPC-122 isolate sufficient determinant and zero-mode exponent ledgers [9, 10]. TPC-123–130 are designed to expose a literal native archive and one arithmetic reduction chain [11–18]. These pieces do not compose merely because their displayed powers look compatible.

There are four independent failure modes.

1. A premise may be a renamed form of the desired conclusion.
2. The same physical event may be charged twice, or two separate occurrences of one formula may be merged and undercharged.
3. An energy exponent may be entered as an amplitude exponent.

4. An arithmetic saving, determinant reserve, or soft-remainder estimate may be recycled as physical endpoint slack.

We prevent these errors before stating the synthesis theorem.

2 Typed route records and three tail namespaces

Definition 2.1 (Literal scope). A route scope is a record

$$\mathfrak{s}_X = (h_0, W, \Omega_X, \nu_X, \mathfrak{a}_X, \mathfrak{q}_X),$$

where $h_0 \neq 0$ is prescribed, W is the fixed physical test weight, Ω_X is the actual active atom set, ν_X is the single physical normalization, $\mathfrak{a}_X$ is a content-addressed archive manifest, and $\mathfrak{q}_X$ records the quantifier range. Two estimates may be composed only after an exact crosswalk identifies all six fields.

The manifest is not cosmetic. A final scalar equality cannot detect a deleted atom paired with an inserted null pair. The path archive and physical reassembly proposed in TPC-123 and TPC-124 are therefore ancestors of every downstream use, not conclusions inferred from the value of the hard packet.

Definition 2.2 (Disjoint tail roles). The certificate language has three different namespaces:

- H3.sqtail : the squarefree-divisor truncation remainder in the arithmetic packet,
- H5.content : the omitted content term in the distinguished zero mode,
- Phys.hitail : the physical high, ultra, boundary, and mask complement.

Two records from these namespaces are never exact duplicates merely because both are called a tail.

Proposition 2.3 (No cross-namespace cancellation). *Without a literal joint theorem, a saving recorded in one namespace of [definition 2.2](#) supplies no bound in either of the other two.*

Proof. The three objects have different domains and roles. The first is an upstream arithmetic remainder inside a raw packet, the second is a zero-mode summand, and the third is a physical outer reconstruction. An inequality for one does not contain either of the other two as a term. Identifying their exponents therefore changes the bounded object. A joint theorem may of course replace the three records, but it must display the common literal expression and list all absorbed dependencies. $\square$

Every complete certificate also carries the exact coverage map

- H3.sqtail $\longrightarrow$ raw packet theorem or one separate arithmetic residual,
- H5.content $\longrightarrow$ the H5 zero-mode admission record,
- Phys.hitail $\longrightarrow$ the complete original-scale soft/physical return.

Each literal tail occurrence must appear exactly once in this map. Different names are not by themselves a coverage proof, and one occurrence may not be reused in two destinations.

3 An acyclic proof DAG with a semantic firewall

Definition 3.1 (Proof record). A proof record is

$$v = (\text{id}, \text{statement}, \text{scope}, \text{source}, \text{parents}, \text{level}).$$

The directed graph has an edge $u \rightarrow v$ when the theorem recorded at v invokes the statement recorded at u . Every source theorem must precede its consumer in a displayed topological order.

Definition 3.2 (Semantic target firewall). Primitive records may not assume any of the following, or an equivalent reformulation:

$$B_{h_0, \delta}(X) = o(X), \quad \text{the target H3 packet saving,} \quad D_X \geq D_X^{\text{target}}, \quad |Z_X| \leq Z_X^{\text{target}}.$$

In addition:

- (i) archive completeness may not be inferred from final scalar agreement;
- (ii) a physical residual may not be called $o(X)$ by invoking the final hard-packet estimate;
- (iii) phase participation may not be defined from D_X/E_X ; and
- (iv) a bilinear reduction may not use H3 itself to bound its residual terms.

Theorem 3.3 (Acyclic certificate composition). *Let $G = (V, E)$ be a finite proof graph whose records have one literal scope. Suppose:*

- (a) G has a topological ordering;
- (b) each nonprimitive record is obtained by a valid invocation of its parent records;
- (c) each primitive record has an independently proved source theorem; and
- (d) the semantic firewall of [definition 3.2](#) holds.

Then the root statement is a noncircular consequence of the primitive records. Conversely, a directed cycle or a target-equivalent primitive premise invalidates this certificate, even if every local algebraic identity is correct.

Proof. Induct along the topological order. Primitive nodes are justified by (c). When all parents of a nonprimitive node have been justified, (b) justifies that node. This reaches the root without using a later node. Condition (d) prevents the conclusion from being inserted as an apparently primitive premise. A directed cycle has no such induction order. A target-equivalent primitive premise makes the implication formally true but circular, which is excluded by definition. $\square$

Remark 3.4 (The firewall is partly semantic). A script can reject forbidden identifiers and graph cycles, but it cannot decide mathematical equivalence of arbitrary theorem statements. The proof-carrying manifest must therefore include a human-audited statement hash and a short independence note for every primitive record.

4 Occurrence-level physical tokens

TPC-118 introduced a nonduplicated physical registry [\[7\]](#). Here we refine its identifier so that a repeated application of one map is not silently merged.

Definition 4.1 (Occurrence token). An occurrence token is

$$\tau = (\text{occ}, \text{map}, \text{domain}, \text{codomain}, \text{scope}, \text{scale}, \lambda, \text{source}, \text{deps}).$$

Here occ is a path-and-stage identifier and $\text{scale} \in \{\mathbf{A}, \mathbf{E}\}$ records amplitude or energy. The exponent λ is unknown until a theorem supplies it. A joint token replaces, rather than supplements, the occurrence identifiers in deps . Occurrence identifiers are globally unique, independently of scale. The dependency graph of joint tokens must be acyclic, and the maximal retained tokens must give a disjoint exact cover of the primitive physical occurrences. Self-dependency, overlapping joint covers, and retaining both a joint token and one of its dependencies invalidate the registry.

Proposition 4.2 (Safe exponent transport). *For upper bounds on a fixed route:*

- (i) sequential amplitude losses add;
- (ii) an energy loss $X^{e+o(1)}$ contributes amplitude loss $e/2$;
- (iii) an ℓ^1 summation over $X^{\mu+o(1)}$ branches contributes μ ;
- (iv) the replacement $\mu \mapsto \mu/2$ requires a proved orthogonality or square-function inequality; and
- (v) two occurrences of the same map remain distinct unless a theorem bounds their literal composition.

Proof. Items (i) and (ii) follow from products of operator norms and taking a square root. Item (iii) is the triangle inequality. Cauchy–Schwarz gives item (iv) only after a square-sum bound is available; branch count alone is insufficient. For (v), applying the same operator at two stages produces a product of two events. Equality of formulas does not identify their occurrences. $\square$

For a valid retained registry $\mathcal{R}$, the canonical amplitude loss is therefore

$$\Lambda_{\text{phys}} = \sum_{\tau \in \mathcal{R}} \begin{cases} \lambda_{\tau}, & \text{scale}(\tau) = \text{A}, \\ \lambda_{\tau}/2, & \text{scale}(\tau) = \text{E}. \end{cases} \quad (1)$$

The conversion in (1) is part of registry aggregation, not a detached numerical example.

Remark 4.3 (Actual composite versus quotient envelope). The quotient condition number of TPC-113 is a sharp uniform envelope, not automatically the norm of the actual H9 composite. Likewise the TPC-114 factor k_{max} is a squared-energy factor and does not automatically equal a max-packet-to-scalar cost. The registry records only the theorem-backed literal composite and its declared norm [2, 3].

5 The determinant–zero ledger is separate

TPC-121 supplies the conditional determinant exponent

$$\lambda_D^{\text{cert}} = (\lambda_E + 2\lambda_{\phi} - \gamma_R)_+,$$

while TPC-122 supplies, when $\delta_{\text{pre}} \geq \ell_Z$,

$$\eta_Z^{\text{cert}} = \min\{\delta_{\text{pre}} - \ell_Z, \eta_{\text{cont}}\}.$$

These are callable only when their literal growing premises have been proved on compatible archived fibers.

Proposition 5.1 (Typed H5 admission test). *The TPC-121/122 certificates pass the determinant–zero compatibility gate precisely when*

$$(\lambda_E + 2\lambda_{\phi} - \gamma_R)_+ \leq 2 \min\{\delta_{\text{pre}} - \ell_Z, \eta_{\text{cont}}\}$$

and $\delta_{\text{pre}} \geq \ell_Z$. The nonnegative number

$$\rho_{\text{det}} = 2 \min\{\delta_{\text{pre}} - \ell_Z, \eta_{\text{cont}}\} - (\lambda_E + 2\lambda_{\phi} - \gamma_R)_+$$

is determinant reserve, not a negative physical token.

Proof. Substitute the two admissible exponents into the exact TPC-112 cone $\lambda_D \leq 2\eta_Z$ [1]. The cone is an admissibility condition between a lower determinant-energy bound and an upper zero-mode bound. Neither quantity is an operator norm in the physical synthesis chain. Hence their difference cannot be subtracted from that chain without a new literal intertwining theorem. $\square$

Remark 5.2 (Comparison-scale invariance). In TPC-122 a rescaling $A_f \mapsto X^c A_f$ changes $(\delta_{\text{pre}}, \ell_Z)$ to $(\delta_{\text{pre}} + c, \ell_Z + c)$. Only $\delta_{\text{pre}} - \ell_Z$ is invariant. Optimizing the two entries separately is exponent bookkeeping, not a stronger zero-mode theorem.

6 The end-to-end synthesis theorem

Let $\mathcal{B}_X^{\text{ns}}$ be the retained nonsoft packet family. Its cardinality is allowed to be $X^{o(1)}$, with every error term below uniform in b . Let $\mathcal{S}_X$ be the literal physical synthesis after all selected route maps.

Theorem 6.1 (Typed fixed-shift endpoint synthesis). *Fix one route scope $\mathfrak{s}_X$. Assume:*

(i) a complete native archive and exact reconnection identity

$$B_{h_0, \delta}(X) = R_{\text{soft}}(X) + \mathcal{S}_X(\{\mathcal{P}_b\}_{b \in \mathcal{B}_X^{\text{ns}}});$$

(ii) $R_{\text{soft}}(X) = o(X)$ after its complete original-scale outer summation;

(iii) for one fixed $\sigma_{\text{raw}} \geq 0$,

$$\max_{b \in \mathcal{B}_X^{\text{ns}}} |\mathcal{P}_b| \leq X^{1-\sigma_{\text{raw}}+o(1)};$$

σ_{raw} is an amplitude saving after every energy-to-amplitude square root and the packet normalization have been executed; a quadratic exponent may not be inserted unchanged;

(iv) the H5 admission test of [proposition 5.1](#), whenever that transfer is used by the selected packet theorem;

(v) an acyclic proof certificate satisfying [theorem 3.3](#); and

(vi) one complete theorem-backed occurrence registry whose canonical loss is (1), and an actual $\ell^\infty(\mathcal{B}_X^{\text{ns}}) \rightarrow \mathbb{C}$ synthesis theorem with

$$|\mathcal{S}_X(\{\mathcal{P}_b\})| \leq X^{\Lambda_{\text{phys}}+o(1)} \max_b |\mathcal{P}_b|.$$

This norm includes the weighted packet census, packet multiplicity, outer coefficients, branch aggregation, and every selected physical occurrence exactly once.

Then

$$\boxed{|B_{h_0, \delta}(X)| \leq o(X) + X^{1-\sigma_{\text{raw}}+\Lambda_{\text{phys}}+o(1)}}.$$

In particular,

$$\boxed{\sigma_{\text{raw}} > \Lambda_{\text{phys}}} \implies B_{h_0, \delta}(X) = o(X).$$

Proof. Apply the exact identity in (i), the triangle inequality, and (vi). Then insert (iii):

$$|B_{h_0, \delta}(X)| \leq |R_{\text{soft}}(X)| + X^{\Lambda_{\text{phys}}+o(1)} X^{1-\sigma_{\text{raw}}+o(1)}.$$

The complete synthesis norm in (vi), rather than the bare statement $|\mathcal{B}_X^{\text{ns}}| = X^{o(1)}$, absorbs the weighted packet census and outer aggregation. Assumptions (iv) and (v) certify that neither an incompatible H5 transfer nor a circular source was used to obtain the two displayed bounds. If $\sigma_{\text{raw}} - \Lambda_{\text{phys}} > 0$, the power term and the soft term are both $o(X)$. $\square$

Corollary 6.2 (MVP1 endpoint). *For the MVP1 raw packet target $\sigma_{\text{raw}} = 1/400$, the strict physical condition is*

$$\boxed{\Lambda_{\text{phys}} < 1/400}.$$

Equality produces only $X^{1+o(1)}$ from the power term and is not a pass.

Proof. Substitute 1/400 in [theorem 6.1](#). At equality the unspecified $o(1)$ has no controlled favorable sign. $\square$

Remark 6.3 (A reduction is not item (iii)). The exact Liouville pullback, squarefree expansion, quantifier firewall, and Fejér four-sign reduction of TPC-127–130 may identify a sufficient arithmetic target. They satisfy (iii) only after the resulting growing masked four-sign forms and every truncation residual have been proved with the required uniform power.

7 Route certificates and first-failure output

For each affine parameter cell r , TPC-118 writes

$$\Lambda_r(x) = c_r^\top x + d_r, \quad A_r x \geq b_r, \quad x \geq 0,$$

after exact dependency replacement [7]. The arithmetic saving σ_r belongs in the right side of the strict comparison, not as a negative physical token.

Proposition 7.1 (Three endpoint outcomes). *For one selected route:*

- (i) *an exact theorem-backed primal point with $\Lambda_r(x) < \sigma_r$ is a strict endpoint pass;*
- (ii) *an exact dual lower certificate with optimum $\Lambda_r^* \geq \sigma_r$ stops that fully specified route; and*
- (iii) *an unknown token, unproved constraint, missing archive, or unavailable upper/lower certificate is INCOMPLETE.*

These states are mutually exclusive. A stop applies only to the selected literal route and allowed replacement menu.

Proof. The first two conclusions are weak duality with threshold σ_r . They cannot both hold. Every remaining state lacks a certificate for either strict upper compatibility or a route lower obstruction and is therefore incomplete. $\square$

The manifest should emit the first node in topological order that is absent, conditional, or scope-incompatible. This “first missing” record is diagnostic; it cannot be replaced by a later consequence.

8 Deterministic regression and present status

The standard-library script `experiments/tpc131_typed_synthesis_audit.py` verifies:

1. an acyclic example and rejection of a deliberate cycle;
2. a primitive-source firewall and rejection of target-as-premise;
3. occurrence-level tokens and exact joint dependency replacement;
4. energy-to-amplitude conversion;
5. nonmerging of the three tail namespaces;
6. the TPC-121/122 compatibility formula;
7. strict, equality, above-threshold, and incomplete endpoint states; and
8. refusal to label a finite identity or reduction as L2.

The numeric values are certificate-format regressions, not estimates of any actual growing TPC coefficient.

Thus TPC-131 does not return GO. It makes the future invocation falsifiable: either every primitive theorem, archive edge, and physical occurrence is present with positive final slack, or the certificate reports the first missing or stopped node.

Table 1: Claim ledger for TPC-131.

Statement	Level	Status
Acyclic composition and semantic firewall	L0	Proved.
Occurrence registry and exponent transport	L0/L1	Proved as a typed certificate calculus.
General $\sigma_{\text{raw}} - \Lambda_{\text{phys}}$ synthesis	Conditional L1	Proved implication.
Actual complete native archive and reconnection	L1 target	Not supplied by the finite regression.
Actual H3/H5 growing exponents	L2 target	Open; no positive exponent is claimed.
Actual strict endpoint and $B_{h_0, \delta} = o(X)$	Global target	Incomplete.

9 Conclusion

The central formula is not a new arithmetic estimate:

$$\text{raw packet saving} - \text{physical occurrence loss} = \text{final fixed-power slack}.$$

Its value is that both terms are now typed and proof-carrying. Determinant compatibility, squarefree truncation, zero-mode content, and physical tails remain in their correct namespaces. Equality at the physical endpoint stops the selected route, while equality in the determinant cone is allowed with zero determinant reserve.

No statement here proves H1–H9, a fixed- h_0 Möbius or Liouville cancellation estimate, a Hardy–Littlewood asymptotic, a prime-pair lower bound, or the twin-prime conjecture. The result is a strict end-to-end interface for deciding when such inputs would actually compose.

References

- [1] Liang Wang. Determinant-zero compatibility at the high-beta endpoint, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-112-determinant-zero-compatibility-audit. TPC-112.
- [2] Liang Wang. Canonical physical frame conditioning and reassembly audit, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-113-canonical-physical-frame-reassembly. TPC-113.
- [3] Liang Wang. Literal residual grouping and the native gram, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-114-residual-grouping-native-gram. TPC-114.
- [4] Liang Wang. Fixed-shift de-aliasing after frozen averaging, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-115-fixed-shift-dealiasing-localization. TPC-115.
- [5] Liang Wang. Complete high-frequency and ultra-tail closure at the original physical normalization, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-116-complete-tail-closure. TPC-116.
- [6] Liang Wang. The tpc-18 full-block physical remainder, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-117-full-block-physical-cover. TPC-117.
- [7] Liang Wang. Strict endpoint optimization for the physical tpc route, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-118-strict-endpoint-optimization. TPC-118.
- [8] Liang Wang. Original hard-packet reconnection, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-119-original-hard-packet-reconnection. TPC-119.
- [9] Liang Wang. A fiber phase-coherence certificate for post-bin determinant energy, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-121-fiber-phase-coherence-certificate. TPC-121.
- [10] Liang Wang. Signed-prefix to zero-mode transfer, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-122-signed-prefix-zero-mode-transfer. TPC-122.
- [11] Liang Wang. Literal native-atom path archives for the tpc-15 hard packet, 2026. TPC-123 manuscript.
- [12] Liang Wang. Provenance-faithful physical reassembly of the tpc hard packet, 2026. TPC-124 manuscript.
- [13] Liang Wang. Prescribed-shift intertwining through the physical archive, 2026. TPC-125 manuscript.

- [14] Liang Wang. A literal fixed-shift generic-affine packet, 2026. TPC-126 manuscript.
- [15] Liang Wang. Determinant-two liouville pullback, 2026. TPC-127 manuscript.
- [16] Liang Wang. Squarefree masks in the shift-two frame, 2026. TPC-128 manuscript.
- [17] Liang Wang. A log-chowla quantifier firewall for the literal fixed-shift route, 2026. TPC-129 manuscript.
- [18] Liang Wang. A Fejér four-sign gate for H3, 2026. TPC-130 manuscript.